

Conditional Residual Clinical Notes Learner

A staged residual architecture for surgical procedure-duration prediction

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September 2026

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Conditional Residual Clinical Notes Learner

A staged residual architecture for surgical procedure-duration prediction from structured records and pre-operative clinical narrative

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Prepared for	Opmed.ai
Document status	Complete internal method record
Model identity	strict_meta_gate_top_refine_oof_selected
Held-out MAE	31.5564180731 minutes

0. How to read this document

This is the complete record of the Conditional Residual Clinical Notes Learner (CRCNL): its motivation, its data handling, every preprocessing decision, the full mathematical formalism of every stage, the risk-residual analysis, the validation and leakage audit, and — importantly — **what has not yet been done**.

Three things deserve emphasis before the detail:

1. **This is a residual architecture, not a monolithic model.** The clinical notes never attempt to predict duration. They predict only what the structured record could not.
2. **We have completed exactly one residual iteration.** The architecture is explicitly designed to be iterated to convergence. Section 15 gives the formal iteration operator, the shrinkage schedule, and four candidate stopping rules. Nothing in the current result exhausts the method.
3. **The narrative extraction layer is the weakest link and is scheduled for replacement.** The current concept extraction is five hand-written regular expressions. Section 8 specifies the replacement: a service-based extraction layer with retrieval, a bi-encoder ranker and a cross-encoder reranker, designed so that the pipeline stops depending on hand-authored lexical patterns.

All mathematics uses Notion-compatible inline delimiters, e.g. y_i , $\hat{y}_i$, MAE.

1. Executive summary

1.1 Headline result

Quantity	Value
Selected model	strict_meta_gate_top_refine_oof_selected
Training cases	10,882
Held-out cases	2,839
Train/held-out patient overlap	0
Training OOF MAE	32.7156839203 min
Held-out MAE	31.5564180731 min
Gain over the strict PLS manifold	+0.2138455833 min
Paired bootstrap CI for that gain	[+0.1001, +0.3259]
Gain over the procedure-family gate	+0.0511042965 min
Paired bootstrap CI for that gain	[-0.0108, +0.1123]
Improved-case rate vs strict PLS	48.89%
Large wins (> 10 min)	23
Large losses (< -10 min)	16

The saved prediction table recomputes independently to 31.556418073067334.

1.2 The staged ladder

Each stage is a strict refinement of the one above it. Every number is held-out MAE in minutes.

Stage	Model	Train OOF MAE	Held-out MAE
0	Zero-information (global median)	—	≈ 81
1a	xgb_tuned_v2 (squared error)	—	32.544887
1b	xgb_absoluteerror_same_shape	—	32.483091
1c	Tabular base $0.6 \cdot (1a) +$ $0.4 \cdot (1b)$	32.966712	32.239636
2	Strict PLS note/tabular manifold	32.796418	31.770264
3	Procedure-family calibration	32.769280	31.607522
4	Risk/residual candidate bank <i>(bank, not a single model)</i>		—
5	Top-refine meta-gate (selected)	32.715684	31.556418

Total improvement from tabular base to final: **0.683218 minutes**, of which **0.469372 minutes** — roughly 69% — comes from the clinical notes alone (stage 1c → stage 2).

1.3 What the notes are worth

The single most important scientific claim in this work:

Pre-operative clinical narrative, restricted to text finalised more than two days before the case, and used **only** as a residual correction on a strong tabular model, is worth **0.469 minutes of MAE** with a clustered bootstrap interval of $[+0.3358, +0.6066]$.

That interval excludes zero. It is the one result in this document that does not depend on any gate, bin, or selection heuristic.

2. Motivation

2.1 The operational problem

An operating-room schedule is assembled from estimates of case duration. The cost structure is asymmetric in practice but linear in the unit that matters:

- An under-estimate runs the day over. Staff are held past shift; later cases are delayed or cancelled.
- An over-estimate leaves the room idle. Capacity is destroyed silently.

Both are paid for in **minutes of error**, and a minute of overrun on an easy case costs the same as a minute on a hard one. This is the argument for absolute error rather than squared error, and it is made precise in Section 3.2.

2.2 Why the procedure code is not enough

The scheduling estimate is normally derived from the procedure code. **Procedure code and procedure difficulty are not the same object.** Two cases carrying identical codes can differ by hours:

- a first-time implant in normal anatomy, versus

- a redo with abandoned leads, venous occlusion, and prior surgical scarring.

That distinction is frequently absent from the structured record. It is frequently present in the pre-operative notes — written in prose, by many hands, over the weeks before the case.

The empirical signature of this is visible in our own family table (Section 13.3): the lead/device extraction family has a base MAE of 49.14 minutes and a mean actual duration of 216 minutes, while the device-implant family has a base MAE of 25.52 minutes at a mean duration of 145 minutes. The hard families are hard in a way the structured record does not see.

2.3 Why residual learning rather than joint learning

Three reasons, in order of importance.

Statistical. The tabular model already captures procedure identity, schedule, provider, demographics and comorbidity structure. A joint note+tabular model must relearn all of it from a text channel that is noisy, high-dimensional and heterogeneous. With $n \approx 11,000$ training cases that is a losing trade. The residual formulation gives the note channel a target that is *already orthogonal* to everything the tabular model explains.

Operational. A residual correction degrades gracefully. If the note channel is unavailable for a case, the prediction falls back to the tabular base with no re-architecture. Note coverage is 92.05% on train and 99.79% on held-out — not 100% — so this matters.

Epistemic. A residual correction is *auditable as a correction*. We can report its mean magnitude (3.36 minutes), its alignment rate with the true residual (56.3%), and its per-family gain. None of that is legible in a joint model.

2.4 Why “conditional”

The correction is not applied uniformly. The architecture learns, at three increasing levels of resolution, *how much to trust the note correction for this case*:

1. by **procedure family** (Section 11);
2. by **learned risk stratum** — where the model expects to be unreliable (Section 12);
3. by **candidate disagreement** — where two already-calibrated predictors disagree (Section 13).

This is the “conditional” in Conditional Residual Clinical Notes Learner. The correction is a function; the *trust* in the correction is a second, separately learned function.

2.5 The central empirical lesson

Stated once here because it shaped the whole architecture:

Error magnitude is learnable. Error direction is not.

Across every risk head we trained, out-of-fold correlation with absolute error reached 0.45–0.46; out-of-fold correlation with signed residual or with final-vs-base gain never exceeded 0.08. The architecture therefore treats risk as a **trust-allocation signal** — deciding how much of a correction to apply — and never as a direction oracle. Section 12.4 gives the numbers.

3. Problem formalism

3.1 Target and estimand

For case i let

$$y_i = \text{case_actual_duration_time}_i \in (0, \infty)$$

measured in minutes. Let x_i denote the structured (tabular) covariates and z_i the eligible pre-operative note bundle. The prediction must be issued at time τ_i , the moment the case is scheduled.

We seek $\hat{f}$ minimising

$$\text{MAE}(f) = \frac{1}{n} \sum_{i=1}^n |y_i - f(x_i, z_i)|.$$

3.2 Absolute loss targets the conditional median

This is not a stylistic preference; it changes what the model learns. For fixed x ,

$$\arg \min_a \mathbb{E}[(y - a)^2 | x] = \mathbb{E}[y | x], \quad \arg \min_a \mathbb{E}[|y - a| | x] = \text{med}(y | x).$$

Proof of the second. Let F be the conditional CDF of Y given x and $g(a) = \mathbb{E}[|Y - a| | x]$. Writing

$$g(a) = \int_{-\infty}^a (a - y) dF(y) + \int_a^{\infty} (y - a) dF(y)$$

and differentiating gives $g'(a) = F(a) - (1 - F(a)) = 2F(a) - 1$, which vanishes exactly at $F(a) = \frac{1}{2}$. g is convex, so this is the global minimum. ■

Consequence for a skewed target. The duration distribution is right-skewed (mean exceeds median by roughly 21 minutes on the parent cohort). Optimising squared error would inflate predictions on the ordinary majority in order to hedge against a rare tail. Absolute loss does not; it fits the typical case and lets the tail be wrong. For a schedule, that is the correct behaviour.

This is why the tabular base is a **blend of a squared-error and an absolute-error learner** rather than either alone (Section 9.3): the squared-error member contributes conditional-mean sensitivity, the absolute-error member contributes median-like robustness, and the 60/40 blend was selected on out-of-fold MAE.

3.3 The irreducible floor

For any information set $\mathcal{G} = \sigma(X, Z)$,

$$R^*(\mathcal{G}) = \inf_{f \text{ } \mathcal{G}\text{-measurable}} \mathbb{E}[Y - f] = \mathbb{E}[\text{MAD}(Y | X, Z)], \quad \text{MAD}(P) := \min_a \mathbb{E}_P |Y - a|.$$

R^* is a property of the joint distribution, not of any model. Its zero-information counterpart is $R^*(\emptyset) = \text{MAD}(Y) \approx 81$ minutes, the reference against which every reported error should be read. A model at 31.56 minutes has removed roughly **61%** of the marginal dispersion.

R^* is not decoration: it is the quantity that makes the stopping rule in Section 15 well posed. An iteration scheme that cannot say how far it is from the floor cannot say when to stop.

3.4 The decomposition the architecture implements

The architecture is a constructive realisation of

$$y_i = \underbrace{m(x_i)}_{\text{structured base}} + \underbrace{g(x_i, z_i)}_{\text{note residual correction}} + \underbrace{\varepsilon_i}_{\text{irreducible or unmodelled}}$$

with the crucial refinement that g is not applied raw but **gated**:

$$\hat{y}_i = m(x_i) + \gamma(c(x_i)) \cdot \text{clip}(g(x_i, z_i), -C_c, +C_c),$$

where $c(\cdot)$ assigns a case to a trust category and $\gamma(\cdot)$, $C_{(\cdot)}$ are learned, non-negative, category-specific trust and cap parameters.

4. Data

4.1 Sources

Source	Rows	Columns	Contents
Tabular case records	14,195	55	perioperative fields, scheduling timestamps, comorbidity block, historical aggregates
Raw clinical notes	3,179,288	4	epic_id, doc_date, doc_type, content_markdown

Notes are matched to cases by patient and then filtered against a case-specific timestamp (Section 5.3). Raw note text is never distributed.

4.2 Tabular schema

Identity and split fields

case_durable_id, patient_durable_id, scheduled_in_room

Case descriptors

case_anesthesia, case_urgency_level, case_surgery_patient_class, case_admission_patient_class, surgeon_durable_id, procedures, case_num_procedures, case_num_panels, case_num_providers, primary_procedure_name, is_redo, prior_same_proc_duration, redo_time_gap_days

Patient descriptors

patient_sex, patient_smoking_status, patient_age, age_group, bmi

Comorbidity block

Coronary artery disease · heart failure · arrhythmia · conduction block · valve disease · cardiac device · vascular disease · cardiac inflammation · ventricular dysfunction · thromboembolism · pulmonary · renal · metabolic · hypertensive · haematologic · neurologic · hepatic · oncologic · psychiatric · musculoskeletal · lifestyle · gastrointestinal · immune · transplant

Scheduling timestamps

scheduled_setup_start, scheduled_in_room, scheduled_out_of_room, scheduled_cleanup_complete

Outcome-time fields — present in the raw table, excluded from predictors

case_actual_patient_in_room, case_actual_duration_time, case_actual_prep_time, case_actual_procedure, case_actual_wrapup_time

Also excluded: patient_birth_date, all_procs_not_performed, case ID, patient ID.

4.3 The explicitly forbidden source

The former 11-variable LLM feature table

data/mayo/mayo_hrs_llm_features_top11.parquet

is **not** an input, a teacher, a label, a distillation target, or a selection source anywhere in this work. Its filename appears in the codebase only inside defensive sentinels and audit scans. The selected run records `forbidden_llm_feature_file_read = false`, and a static code scan confirms zero access-like hits.

This constraint exists because an earlier result (≈ 32.29) was found to have used that file as a teacher/label source, making it non-independent. Every number in this document is from the post-constraint rebuild.

4.4 Where the data lives

No patient data travels with this document or with the code repository. The source files are held in S3 and must be provisioned in an authorised environment.

Bucket opmed-integration-raw-data-mayo, **prefix** clinical_notes/anonymized_hrs_notes_shared/

File	S3 URI
Dataset — tabular + lags + target	s3://opmed-integration-raw-data-mayo/clinical_notes/anonymized_hrs_notes_shared/mayo_
Anonymized notes	s3://opmed-integration-raw-data-mayo/clinical_notes/anonymized_hrs_notes_shared/mayo_
LLM features (top 11) — not used by this model	s3://opmed-integration-raw-data-mayo/clinical_notes/anonymized_hrs_notes_shared/mayo_

How the files join

```
dataset.case_durable_id    -> LLM features.case_durable_id    (case-keyed, left join)
dataset.patient_durable_id -> notes.epic_id                (1:many, patient-keyed)
```

Notes are keyed by **patient**, not by case. A patient with several procedures has all their notes in one bucket, so attaching a case to “its” notes is a **date filter**, not a join key — which is precisely why the cutoff rule of §5.3 is load-bearing rather than cosmetic.

Notes schema: `epic_id`, `doc_date`, `doc_type`, `content_markdown`.

4.5 A cutoff discrepancy worth recording

The shipped 11-feature LLM table was built with

```
cutoff = case["case_actual_patient_in_room"] - pd.Timedelta(days=2)
```

— anchored to the **actual** in-room time. Every note used is still genuinely dated before the surgery, so this is not leakage from the outcome itself. The defect is subtler: the actual surgery date is not fixed at prediction time, so a case rescheduled later inherits a larger note history than a creation-time deployment would ever have seen.

This model uses `scheduled_in_room - 2 days` instead (§5.3), which is known at prediction time. The data guide reaches the same conclusion independently: *if you are extracting new note-derived features of your own, use `scheduled_in_room` — it is known at prediction time, so it will not have this problem.*

Two consequences follow, and they are the reason the two artefacts must not be mixed:

- Reproducing the shipped 11 features **exactly** requires the actual-time cutoff. A “cleaner” cutoff yields different eligible notes and therefore different feature values.

- This model is built on the clean cutoff and **excludes those features entirely**, so the two are not interchangeable and the difference is not a bug in either.

4.6 A different cohort design also exists

The shared data guide describes an **expanding-window quarterly CV** on the actual case date: warm-start 2022 Q3-Q4 (lag pools only, never trained or evaluated), train-only 2023 Q1-Q4, eight OOF validation folds over 2024 Q1 – 2025 Q4, holdout 2026 Q1+.

This model instead uses a single temporal cut at the 80th percentile of `scheduled_in_room` with patient disjointness (§5.2). The quarterly design is closer to the nested forward-time selection recommended in §15.7 as the next validation step, and adopting it is a concrete, well-specified upgrade rather than an open research question.

Two dataset caveats from the same guide, neither of which affects the figures here but both of which matter on a rebuild: 6 rows have a null `primary_procedure_name` and 23 have `all_procs_not_performed = True` — the training pipeline filters only on target validity and does **not** drop these. The target is 100% populated and positive as shipped.

5. The prospective information boundary

5.1 Filtration

Let $\mathcal{F}_t$ be the σ -field of everything knowable at time t . A feature is admissible for case i if it is $\mathcal{F}_{\tau_i}$ -measurable, where τ_i is the scheduling moment.

Leakage is a measurability failure, not a statistical subtlety. It is a mis-specified σ -field, and it produces error estimates that cannot be attained in deployment.

The diagnostic value of the exclusion in Section 4.2 is worth stating as a standing rule:

With the three `case_actual_*` summands included, a routine gradient-boosted model reaches an apparent MAE of 2.56 minutes against a zero-information baseline of 81. **Any reported error more than an order of magnitude below the marginal dispersion should be treated as a leak until proven otherwise.**

5.2 Case filtering and the temporal patient-disjoint split

Rows without `scheduled_in_room`, without the target, or with non-positive duration are removed. Remaining rows are sorted by scheduled time.

Let t_i be the scheduled in-room timestamp and $q_{0.8}$ its 80th percentile. Candidate held-out patients are those with any case at or after $q_{0.8}$. Then:

$$\begin{aligned} \text{train} &= \{i : t_i < q_{0.8} \wedge \text{patient}(i) \notin \mathcal{P}_{\text{held-out}}\}, \\ \text{held-out} &= \{i : t_i \geq q_{0.8} \wedge \text{patient}(i) \in \mathcal{P}_{\text{held-out}}\}. \end{aligned}$$

Item	Value
Train cases	10,882
Held-out cases	2,839
Train patients	8,948
Held-out patients	2,499
Patient overlap	0
Missing patient IDs	0 / 0

Two separate controls are at work and they answer different questions. The **temporal** cut answers *how would this have performed if fielded on the boundary date*. The **patient disjointness** prevents the model from memorising a patient whose entire note history and comorbidity documentation repeats across cases — without it the evaluation measures recall rather than generalisation.

5.3 The note cutoff

A note is eligible for case i only if

$$\text{doc_date} < \text{scheduled_in_room}_i - 2 \text{ days.}$$

Two design points matter and are easy to get wrong.

Anchor on the scheduled time, not the actual time. The actual in-room time is unknown at prediction. Anchoring on it would let a case rescheduled from March to June inherit three months of note history no deployment could have seen.

Use a two-day margin, not zero. A note written the morning of the procedure is technically pre-operative but is unavailable at scheduling and may encode the day’s operative plan in a form close to the outcome itself. Two days removes that class of near-outcome documentation.

Note-cache construction settings

Setting	Value
Timestamp policy	exact timestamp
Same-day pre-operative note types	disabled
Post-operative document types	rejected
Maximum lookback	548 days
Maximum notes per case	24
Maximum characters per note	2,200
Maximum characters per case	48,000
Maximum selected sentences	72

Audit of the built cache

Audit item	Value
Raw note rows scanned	3,179,288
Rows matched by patient	3,144,540
Post-operative doc-type rows rejected	145,426
Eligible note events attached	295,463
Cases with a bundle	13,721
Cases with at least one used note	12,850
Note dates checked	295,463
Cutoff violations	0
Smallest margin before cutoff	2.5 hours
Train note coverage	92.0511%
Held-out note coverage	99.7887%
Median selected notes per case	24 (both splits)
Median characters per case	24,377 train / 24,702 held-out

A nuance worth recording. Historical notes can legitimately contain words such as *post* or *discharge*, because they may describe an older event. **Timestamp eligibility, not a word blacklist, is the decisive boundary.** As a secondary check, the high-signal bundles contained zero actual duration probe hits on either split and total-time language remained uncommon. Production auditing should retain both the timestamp check and the lexical probe.

5.4 The scheduling-feature caveat

The run uses *planned* scheduling information. Derived fields:

- `scheduled_year`, `scheduled_month`, `scheduled_dayofweek`, `scheduled_hour`
- `scheduled_is_weekend`
- `scheduled_room_minutes` = scheduled out-of-room — scheduled in-room
- `scheduled_setup_lead_minutes` = scheduled in-room — scheduled setup start
- `scheduled_cleanup_minutes` = scheduled cleanup complete — scheduled out-of-room

These are **not** actual outcome times, and they are knowable at τ_i by construction — the booked slot exists *because* the case was scheduled. But they change the prediction task. The 31.5564 result applies to the **schedule-informed** setting. If the deployment question is *predict before a schedule is built*, these fields must be removed and the entire pipeline retrained and re-evaluated.

Two residual risks attach to this and should be audited before deployment:

1. **Revision risk.** If the scheduling system overwrites the booked slot when a case is amended, the stored field is not $\mathcal{F}_{\tau_i}$ -measurable. The check is whether the source keeps a booking-version history or an `as_of` timestamp; failing that, an anomalously tight ridge along the diagonal of booked-vs-actual is the diagnostic signature.
2. **Feedback risk.** The booked slot may *causally* influence realised duration — staff pace to the block. Deploying a model to *set* the slot then changes the distribution it learned. This is a deployment-loop concern, not leakage, but it belongs in the limitations.

6. Tabular preprocessing and data handling

6.1 Feature frame construction

The builder removes, in order: identity columns, `patient_birth_date`, the target, every column beginning with `case_actual_`, date-suffixed columns, and the raw schedule timestamps **after** deriving the permitted planned-time features of Section 5.4.

Then:

- Boolean variables are cast to integers.
- Numeric missing values are imputed with the **training median**.
- Categorical missing values are imputed with the **most frequent training category**.
- Categories are one-hot encoded, unknown categories ignored, with a minimum category frequency of 10 where supported.
- An explicit `bmi_missing` indicator is added, because BMI missingness is both common and informative.

Every one of these is a *learned* quantity and every one is estimated on the fitting partition only. Computing them on pooled data is transductive leakage: the evaluation rows influence the representation even though their labels are unused.

6.2 Smoothed target encodings

Two high-cardinality categoricals — `primary_procedure_name` and `surgeon_durable_id` — are target-encoded with additive smoothing. For a category c with count n_c , category mean $\bar{y}_c$, global mean $\bar{y}$, and smoothing constant k :

$$\text{TE}(c) = \frac{n_c \bar{y}_c + k \bar{y}}{n_c + k}.$$

Field	k
primary_procedure_name	20
surgeon_durable_id	15

Unknown validation or held-out categories receive the training global mean $\bar{y}$.

The fold discipline is the whole point. For each training fold, encodings are fit on the fold-training rows only and *transformed* onto the validation rows. For the held-out split, encodings are fit on the full training split only. Without this, the encoding carries the row's own label and the entire downstream residual construction is contaminated.

7. Note extraction as currently implemented

This section documents what exists. Section 8 specifies what should replace it.

7.1 Bundle assembly

For each case, eligible notes (Section 5.3) are concatenated, capped, and stored as a single `note_text` field per case, together with metadata: note count, character count, document-type counts, recency distribution.

7.2 Lexical representation

A word-level clinical TF-IDF model:

```
TfidfVectorizer(
    lowercase=True,
    stop_words="english",
    token_pattern=CLINICAL_TOKEN_PATTERN,    # r"(?u)\b(?=[^\W\d_]*[^\W\d_])\w[\w-]{1,}\b"
    ngram_range=(1, 2),
    min_df=3,
    max_df=0.97,
    max_features=50000,
    sublinear_tf=True,
)
```

The clinical token pattern deliberately requires at least two non-digit word characters, which suppresses pure numerics and single characters while retaining hyphenated clinical compounds.

With sublinear term frequency, for term j in bundle i with raw count $c_{ij} > 0$:

$$\text{tfidf}_{ij} = (1 + \log c_{ij}) \cdot \log \frac{N + 1}{\text{df}_j + 1},$$

subject to the library's ℓ_2 row normalisation.

7.3 Dimensionality reduction

TruncatedSVD($n = 160$) $\rightarrow$ StandardScaler

fit on training bundles and applied unchanged to held-out bundles. This yields the note latent view $Z \in \mathbb{R}^{n \times 160}$.

7.4 Hand-authored concept indicators — the component to be replaced

Five binary indicators are appended, computed by regular-expression search over the lowered bundle text:

Concept	Patterns
redo_prior_ablation	\bredo\b, repeat\s+ablation, prior\s+ablation
vt_pvc_instability	\bvt\b, \bvf\b, ventricular\s+tachycardia, icd\s+shock
device_lead_complexity	lead\s+(revision\ extraction\ failure\ fracture), generator\s+change
congenital_complex_anatomy	\bfontan\b, transposition, congenital
access_support	\becmo\b, epicardial, pericardial\s+access

Formally, for concept k with pattern set $\mathcal{P}_k$:

$$\text{flag}_{ik} = \mathbb{1} \left[\exists p \in \mathcal{P}_k : p \text{ matches text}_i \right].$$

These are generated from the eligible note text itself and are **not** the forbidden 11-variable LLM table.

7.5 Why this component is the weak link

Four structural problems, each independently sufficient to motivate replacement.

Binary saturation. With bundles of median 24,377 characters, the probability that *some* note mentions congenital at least once is high regardless of whether the current case is congenital. Prevalence bears this out: vt_pvc_instability fires on 40.75% of cases. A flag that fires on two cases in five carries little conditional information.

No negation or attribution handling. no evidence of VT and recurrent VT despite amiodarone produce the identical feature value. Clinical prose is dense with negation, hypotheticals, family history, and historical resolution; a substring match sees none of it.

Brittleness to phrasing and site. The pattern list encodes one institution’s dictation habits. lead revision matches; revision of the RV lead does not. Transfer to another site is unmeasured and probably poor.

Empirically thin. The concept-flag ablation table (Section 13.5) shows *negative* gain lift for four of the five flags — cases where the flag is present improve **less** than cases where it is absent. These indicators are not carrying their weight.

Their measured contribution is small enough that the architecture does not depend on them — which is exactly the right moment to replace them with something better.

8. Proposed replacement: retrieval-based extraction with ranker and reranker

This section is a **design specification, not a completed experiment**. It is written to be implementable and to be evaluated under the same leakage rules as everything else.

8.1 Design goals

Goal	Rationale
Remove dependence on hand-authored lexical patterns	Section 7.5
Handle negation, attribution and temporality	Clinical prose requires it
Produce graded evidence, not binary flags	Binary saturates at bundle length
Remain auditable	A clinician must be able to see <i>which sentence</i> drove a feature
Stay inside the note-eligibility boundary	Non-negotiable
Degrade gracefully if the service is unavailable	Coverage is not 100%

8.2 Architecture

eligible note bundle (per case)

1. SEGMENT → sentence / passage units with doc_type + doc_date
2. RETRIEVE → bi-encoder recall: top-M passages per concept query (M ≈ 50, cosine over a clinical embedding space)
3. RERANK → cross-encoder: (concept query, passage) → relevance keep top-K (K ≈ 8) per concept
4. EXTRACT → structured-output API call over the K passages only → graded assertion with negation / temporality
5. AGGREGATE → per-concept evidence score → feature vector

The critical structural property: **the extraction model never sees the whole bundle**. It sees at most K reranked passages per concept. This bounds cost, bounds context dilution, and makes every produced feature traceable to specific sentences.

8.3 Stage 1 — segmentation

Split each bundle into passages $\{s_{i1}, \dots, s_{iM_i}\}$ of one to three sentences, each carrying its doc_type, doc_date, and character offset. Passages inherit the eligibility of their parent note; the two-day cutoff is enforced at bundle-construction time and never re-litigated downstream.

8.4 Stage 2 — bi-encoder retrieval (the ranker)

Let $\phi(\cdot) \in \mathbb{R}^d$ be a clinical sentence encoder and let q_k be a natural-language query for concept k (for example, “*prior catheter ablation of atrial fibrillation that failed or recurred*”). Rank passages by

$$\text{sim}(q_k, s_{ij}) = \frac{\langle \phi(q_k), \phi(s_{ij}) \rangle}{\|\phi(q_k)\| \|\phi(s_{ij})\|},$$

and retain the top M .

Two properties make this the right first stage. It is **cheap** — passages are embedded once per case and reused across all concepts, so cost is $O(M_i)$ embeddings, not $O(M_i \times K_{\text{concepts}})$. And it is **recall-oriented** — the bi-encoder’s job is not to be right, only to not lose the relevant passage before the reranker sees it.

8.5 Stage 3 — cross-encoder reranking

The bi-encoder scores query and passage independently, so it cannot model their interaction. The cross-encoder scores them jointly:

$$\rho(q_k, s_{ij}) = \sigma\left(\text{CE}([q_k; s_{ij}])\right) \in (0, 1),$$

and the top K passages by ρ are passed forward. This is the stage that distinguishes “*history of VT*” from “*no history of VT*” — a distinction invisible to both regex and cosine similarity.

The staged ranker→reranker structure is standard in retrieval for exactly this reason: recall is cheap and precision is expensive, so buy recall broadly and precision narrowly.

8.6 Stage 4 — extraction via structured service call

For each concept k and case i , issue **one** structured-output call over the K reranked passages, returning a typed object:

```
{
  "concept": "redo_prior_ablation",
  "present": true,
  "assertion": "affirmed",           // affirmed | negated | hypothetical | family_history
  "temporality": "historical",       // current | historical | planned
  "severity": 0.72,                  // graded, 0-1
  "confidence": 0.88,
  "evidence_passage_ids": [4, 17]
}
```

Three engineering constraints make this safe and reproducible:

1. **Determinism.** Temperature zero, pinned model version, schema-validated output. The model version is recorded in the run config and is part of the artefact’s identity.
2. **Caching.** Keyed on (case_id, concept, passage_id_set, prompt_version, model_version). Re-runs cost nothing; a prompt change invalidates cleanly.
3. **Fallback.** If the service is unavailable or the schema validation fails, fall back to the Section 7.4 regex flag and set an extraction_degraded indicator. The model must never silently change its input distribution.

Leakage discipline for this stage. The extraction call sees *only* eligible passages. It never sees the target, the case duration, any outcome field, or any other case. It is a deterministic function of admissible text, so it is $\mathcal{F}_{\tau_i}$ -measurable by construction — the same argument that licenses TF-IDF.

8.7 Stage 5 — aggregation into features

For concept k , combine the extraction with the reranker evidence:

$$e_{ik} = \underbrace{\mathbb{1}[\text{assertion} = \text{affirmed}]}_{\text{negation-aware}} \cdot \underbrace{\text{severity}}_{\text{graded}} \cdot \underbrace{\text{confidence}}_{\text{calibration}} \cdot \underbrace{w(\Delta t_{ik})}_{\text{recency}},$$

with an exponential recency weight over the age of the supporting evidence,

$$w(\Delta t) = \exp(-\Delta t / \tau_{\text{half}}),$$

τ_{half} selected inside training folds. Alongside e_{ik} , retain:

- $\rho_{ik}^{\max} = \max_j \rho(q_k, s_{ij})$ — peak retrieval evidence;
- $\bar{\rho}_{ik}$ over the top K — evidence breadth;

- the count of affirmed passages — corroboration;
- an `extraction_degraded` flag.

This yields roughly 4–5 features per concept instead of one binary. With a concept set of 12–15 (extending the current five toward the clinical families that matter), the narrative concept block grows from 5 to ≈ 60 graded features — while becoming *more* auditable, not less, because every value traces to named passages.

8.8 How to prove it is better

Replacing a component is a hypothesis, not an improvement. The evaluation must be pre-declared:

1. Freeze everything downstream of the note representation.
2. Rebuild the note latent view with the new concept block in place of the old.
3. Re-run the full stage-2 manifold and report train-OOF MAE.
4. **Accept only if train OOF improves**, with the held-out set untouched until the decision is made.
5. Report per-family and per-coverage-stratum gain, not only the aggregate.

An honest prior from adjacent evidence: broader *learned* text representations have repeatedly **failed** to help on this problem. Adding 48 SVD components of hashed note text cost 0.62 minutes in a parallel study; adding 100 note-text SVD components raised best-OOF MAE from 32.87 to 33.82 in another. Raw note variance is dominated by *which procedure this is* — information the structured record already holds perfectly — so broader text features contribute redundancy and dilution rather than signal.

This is precisely why the proposed replacement is narrow and targeted rather than broad. It does not add vocabulary; it adds *assertion-level resolution* to a small, clinically motivated concept set. That is a different axis, and it is the axis where the current pipeline is demonstrably weakest.

8.9 Cost and deployment implications

The current pipeline’s headline operational virtue is that prediction requires no GPU, no network call, and no embedding service — a few hundred regular-expression evaluations plus tree traversals. **The proposed design gives that up.** The trade must be made explicitly:

	Current	Proposed
Prediction-time dependency	none	embedding service + extraction API
Per-case cost	negligible	K_{concepts} cached calls
Latency	milliseconds	seconds (cacheable, and pre-computable at scheduling time)
Auditability	pattern list	passage-level evidence
Negation / temporality	none	explicit
Cross-site transfer	untested, likely poor	query-based, expected better

The mitigating fact is that duration prediction is not a real-time task. The prediction is required when the case is *scheduled*, typically days in advance, so extraction can run asynchronously and be cached long before the prediction is needed.

9. Stage 1 — the tabular foundation

9.1 Role

The tabular base estimates broad procedural duration. It must be strong, because everything above it is a *correction*: a weak base leaves the note channel relearning structured information it should never have to see.

9.2 The paired XGBoost models

Two models share an identical tuned tree shape and an identical feature matrix, differing only in objective.

Parameter	Value
n_estimators	1213
max_depth	8
learning_rate	0.013721653065680284
subsample	0.8431891099718941
colsample_bytree	0.5331833452208407
min_child_weight	9.083480959214418
reg_alpha	6.006000053619992e-07
reg_lambda	0.001809354462444587
gamma	1.0186857689274291
max_bin	212
tree_method	hist
eval_metric	mae

Model	Objective	Held-out MAE
xgb_tuned_v2	reg:squarederror	32.544887
xgb_absoluteerror_same_shape	reg:absoluteerror	32.483091

9.3 The 60/40 blend

$$\hat{y}_i^{\text{base}} = 0.60 \hat{y}_i^{\text{sq}} + 0.40 \hat{y}_i^{\text{abs}}.$$

Metric	Value
Train OOF MAE	32.966712
Held-out MAE	32.239636

Note that the blend beats **both** members on held-out (32.2396 vs 32.4831 and 32.5449). The two objectives make genuinely different errors: the squared-error member is sensitive to the conditional mean and therefore to the tail; the absolute-error member is median-like and outlier-resistant. Their convex combination retains some of each.

9.4 Cross-fitting

Within the training cohort, predictions use five-fold GroupKFold with **patient ID** as the grouping variable. A patient never crosses the fit/validation boundary of the same fold.

This produces $\hat{y}^{\text{base, OOF}}$ on train — every entry from a model that never saw that row. Separately, both models are refit on the entire training split and applied to held-out, giving $\hat{y}^{\text{base, test}}$.

Recorded limitation. These internal folds are patient-disjoint but **not chronological**. They control repeated-patient dependence, which is the dominant contamination risk, but they do not simulate forward-time deployment. The top-level held-out split is chronological; the internal estimator is not. A stronger design uses nested forward-time folds with patient purging, and this is the first recommendation in Section 15.6.

10. Stage 2 — the strict clinical-note / tabular shared manifold

This is the stage that produces the largest single gain in the entire system: +0.469372 minutes.

10.1 The residual target

For each training case,

$$r_i = y_i - \hat{y}_i^{\text{base, OOF}}.$$

Using **OOF** base predictions here is essential and not a technicality. An in-sample base residual would be artificially small and structured by overfit; the note head would then learn to undo the base model’s memorisation rather than to add information.

10.2 The two latent views

Note view. From Section 7: TF-IDF $\rightarrow$ TruncatedSVD(160) $\rightarrow$ StandardScaler, plus the concept indicators. Call the result $Z \in \mathbb{R}^{n \times d_Z}$.

Tabular view. The preprocessed tabular matrix is reduced by TruncatedSVD(96), scaled from training statistics, and the **first 64 coordinates** are supplied to the manifold. Call this $X \in \mathbb{R}^{n \times 64}$.

Both reductions are fit on training rows and applied unchanged to held-out rows.

10.3 Partial least squares — the shared manifold

PLS seeks paired directions that maximise **covariance** between the two views. For component h , it solves

$$(w_h, c_h) = \arg \max_{\|w\|=\|c\|=1} \text{cov}(X_{h-1}w, Z_{h-1}c)^2,$$

equivalently the leading singular pair of the cross-covariance matrix $X_{h-1}^\top Z_{h-1}$, followed by deflation:

$$\begin{aligned} t_h &= X_{h-1}w_h, & u_h &= Z_{h-1}c_h, \\ X_h &= X_{h-1} - t_h p_h^\top, & Z_h &= Z_{h-1} - u_h q_h^\top, \end{aligned}$$

with loadings $p_h = X_{h-1}^\top t_h / (t_h^\top t_h)$ and $q_h = Z_{h-1}^\top u_h / (u_h^\top u_h)$.

Why covariance and not correlation. Canonical correlation analysis maximises correlation and is notoriously unstable in high dimension — it will happily find a perfectly correlated pair of near-noise directions. PLS maximises covariance, which is correlation weighted by the variances of both views, and is therefore biased toward directions that actually carry energy. With $d_Z = 160+$ and $n \approx 11,000$ this stability matters.

Fitted configuration. PLSRegression(n_components=40, max_iter=1000) is fit on the 64-dimensional tabular view against the note view. The run config permitted up to 56 components;

the evaluated grid for the surgical profile ended at 40. The selected slice retains the **first 26** components.

For each case this gives a tabular score $t_i \in \mathbb{R}^{26}$ and a note score $u_i \in \mathbb{R}^{26}$ — *what the structured record says*, and *what the narrative says*, expressed in the same 26-dimensional coordinate system.

10.4 The delta geometry

The residual feature vector is

$$\phi_i = [u_i - t_i, |u_i - t_i|, t_i \odot u_i] \in \mathbb{R}^{78},$$

where $\odot$ is the elementwise (Hadamard) product. That is $26 + 26 + 26 = 78$ features.

Each block answers a different question:

Block	Dimension	Meaning
$u_i - t_i$	26	Directional disagreement — do the notes push along or against the tabular expectation, and on which latent axis?
$ u_i - t_i $	26	Magnitude of disagreement , direction-free — how far apart are the two views at all?
$t_i \odot u_i$	26	Alignment or opposition — do the two views agree in sign on this axis?

This is the conceptual core of the whole model. The notes are not used to predict duration. They are used to measure **the geometry of narrative disagreement with the structured expectation**, and that disagreement is turned into a duration correction.

The empirical attribution (Section 13.4) confirms the blocks are not redundant: mean absolute contribution is 8.30 minutes from the directional block, 5.05 from the absolute block, and 2.77 from the product block.

10.5 The robust residual head

$$\hat{r}_i = h(\phi_i), \quad h = \text{StandardScaler} \circ \text{SGDRegressor}.$$

```
SGDRegressor(
    loss="huber",
    penalty="elasticnet",
    alpha=1e-4,
    l1_ratio=0.05,
    max_iter=4000,
    tol=1e-4,
    average=True,
    random_state=seed,
)
```

The Huber loss with threshold δ ,

$$\ell_\delta(e) = \begin{cases} \frac{1}{2}e^2, & |e| \leq \delta, \\ \delta(|e| - \frac{1}{2}\delta), & |e| > \delta, \end{cases}$$

is quadratic near zero and linear in the tails. Since the residual target r_i inherits the duration distribution’s heavy right tail, a squared-error head would be dragged by a handful of extreme cases — precisely the cases where the note signal is least reliable. Elastic-net regularisation with ℓ_1 ratio 0.05 is mostly ridge, appropriate for 78 correlated latent features.

The head is deliberately simple. The representational work is done by the PLS manifold; the head’s job is to map 78 geometric coordinates to one scalar without overfitting. A complex head here consistently underperformed.

Residual OOF predictions use GroupKFold(5) on patient; the head is then refit on the full training split and applied to held-out.

10.6 The trust scalar γ

$$\hat{y}_i^{\text{PLS}} = \hat{y}_i^{\text{base}} + \gamma \hat{r}_i,$$

with γ selected **on training OOF MAE only**, over the grid $\{0.000, 0.025, 0.050, \dots, 1.600\}$.

Selected: $\gamma = 1.550$.

That $\gamma > 1$ is informative. The Huber head, regularised and fit on a heavy-tailed target, systematically *under-shoots* the residual magnitude; γ corrects that shrinkage. It is a calibration scalar, not a confidence statement.

10.7 Stage 2 result

Model identity: pls_raw_word160_26_delta__sgd_huber__g1.550

Metric	Value
Train OOF MAE	32.7964177885
Held-out MAE	31.7702636563
RMSE	47.0862301387
R^2	0.7537433190
Median absolute error	21.3969602483
P90 absolute error	69.8267318237
P95 absolute error	93.2294053259
Within 30 min (W30)	0.6220500176
Within 60 min (W60)	0.8622754491
Mean prediction – actual	–2.8542162009
Under-estimation rate	0.4825642832
Over-estimation rate	0.5174357168
Gain over tabular base	+0.4693722183
Clustered bootstrap CI for that gain	[+0.3357543566, +0.6065614085]
Clustered bootstrap CI for the MAE	[30.6139184325, 33.0492553314]

Run configuration: seed 4242, 5 folds, surgical profile, 127 candidate variants evaluated, 200 bootstrap replicates, forbidden-file read false.

11. Stage 3 — procedure-family calibration

11.1 The gate

The same PLS correction is not equally reliable for every procedure family:

$$\hat{y}_i^{\text{family}} = \hat{y}_i^{\text{base}} + \gamma_{f(i)}(\hat{y}_i^{\text{PLS}} - \hat{y}_i^{\text{base}}),$$

where $f(i)$ is the procedure family and $\gamma_f \geq 0$ is a family-specific trust scalar learned **inside outer patient-group folds**. Small families fall back to the global value.

11.2 Fitted trust scalars

Procedure family	γ_f	Reading
VT/PVC ablation	2.500	correction strongly under-applied
AF/PVI ablation	1.700	correction under-applied
Lead/device extraction-removal	1.525	correction under-applied
Other ablation	1.025	roughly calibrated
LAA closure	1.000	at the global fallback
Device implant	0.775	correction over-trusted
Device revision-generator-upgrade	0.775	correction over-trusted
Diagnostic / testing	0.300	correction largely distrusted
Other	0.275	correction largely distrusted
<i>global fallback</i>	1.000	

The pattern is clinically coherent, which is the strongest argument for it. The families where the note correction is *under*-applied are exactly the complex-substrate ablation and extraction families — the ones where the structured record is least descriptive of difficulty and where narrative context (prior failed ablation, scar burden, venous occlusion) carries real information. The families where it is *over*-trusted are short, standardised, high-volume procedures whose duration is close to deterministic given the code.

11.3 Held-out effect by family

Family	Gain (min)	Improved rate
AF/PVI ablation	+0.5819	58.20%
Lead/device extraction-removal	+0.2409	43.42%
Diagnostic / testing	+0.2296	54.94%
Device revision-generator-upgrade	+0.0816	55.88%
VT/PVC ablation	+0.0262	50.85%
Device implant	-0.0403	48.74%
Other	-0.0663	47.83%

11.4 Result and the VT/PVC warning

Metric	Value
Train OOF MAE	32.7692796199
Held-out MAE	31.6075223696
Gain vs strict PLS	+0.1627412868
95% paired bootstrap CI	[+0.0565, +0.2664]
Improved-case rate	51.32%

AF/PVI is the clearest family-level win. VT/PVC is the clearest hazard. Its $\gamma = 2.5$ is the largest in the table and produces both large wins and large losses; its net gain is only +0.0262. This family needs either a cap (which Section 12 introduces) or a genuine direction signal before it can be pushed harder.

12. Stage 4 — the risk and residual candidate bank

12.1 The question

Where is the current predictor unreliable, and can that be learned *without* using the row's own outcome?

12.2 Risk targets

Four families of learnable target, all computed from OOF predictions:

Target	Definition	What it asks
Absolute error	$ y_i - \hat{y}_i^{\text{OOF}} $	how wrong will we be?
Signed residual	$y_i - \hat{y}_i^{\text{OOF}}$	in which direction?
Tail indicator	$\mathbb{1}(y_i - \hat{y}_i > 30)$	will this be a blow-up?
Gain	$ y_i - \hat{y}_i^{\text{base}} - y_i - \hat{y}_i^{\text{final}} $	did the correction help?

Risk learners: Extra Trees and histogram gradient boosting, over safe tabular features, note meta-data, the base/final/correction triple, and PLS-disagreement geometry.

12.3 Nesting — the part that is easy to get wrong

A risk score is **derived from the target**. If a row's own error contributes to the model that assigns that row a risk category, the category is contaminated and every downstream gate trained on it is optimistic.

The discipline is therefore:

1. Outer patient GroupKFold.
2. For each outer validation fold, the risk model is trained **without** that fold.
3. Score-bin quantile edges are computed from the outer-training scores **only**.
4. Category trust scalars and caps are tuned on the outer-training rows **only**.
5. Held-out outcomes are used for final scoring **only**.

An earlier broad risk grid was held-out-safe but not strictly nested at the OOF-bin-selection step. Only the strictly nested results are reported as claims.

12.4 What the risk heads actually learned

Score model	OOF correlation with its target
risk_abs_extra_trees vs absolute error	0.4569
risk_abs_hgb vs absolute error	0.4519
(earlier pass) risk model vs absolute final error	0.454 – 0.456
gain_hgb vs final-vs-base gain	0.0823
signed_resid_hgb vs signed residual	0.0793
tail30_hgb vs tail indicator	0.0318

This is the table that determined the architecture. A correlation of 0.45 with error *magnitude* is a usable trust signal. A correlation of 0.03–0.08 with error *direction* is not a usable correction signal. Consequently the risk heads are wired as gates on an existing correction, never as correctors in their own right.

12.5 The capped-correction candidate form

The generic candidate in the bank is

$$\hat{y}_i = \hat{y}_i^{\text{anchor}} + \text{clip}(\gamma_c \hat{r}_i, -C_c, +C_c),$$

with $\gamma_c \geq 0$ and $C_c > 0$ tuned per category c (procedure family, learned risk stratum, or their interaction) inside outer folds, subject to a minimum region size and a global fallback.

The cap is what makes a large γ survivable. A family like VT/PVC can be assigned high trust *and* a bound on how far that trust is allowed to move a single prediction.

12.6 Candidate results

Candidate	Train OOF MAE	Held-out MAE	Gain vs strict PLS	95% CI
crossfit_residual_hgb	32.724460	31.737202	+0.033062	—
family_cap_gamma	32.757110	31.628808	+0.141456	[+0.0317, +0.2491]
nested_risk_abs3_x_family_cap_gamma	32.719677	31.607452	+0.162811	[+0.0269, +0.2985]
risk3_x_pls_gap_gamma_pos	32.768101	31.700691	+0.069573	[−0.0476, +0.1844]
risk_abs_hgb4_x_family_cap_gamma	32.800344	31.579011	+0.191253	[+0.0691, +0.3161]
procedure_family_gamma_pos (Stage 3)	32.769280	31.607522	+0.162741	[+0.0565, +0.2664]

Two observations of method, both uncomfortable and both worth recording.

crossfit_residual_hgb won on train OOF and lost on held-out. Best-by-OOF is not a guarantee; it is only the least-biased selector available.

The best strictly nested risk gate ties the simple family gate, at 31.6074524935 versus 31.6075223696 — a difference of 0.00007 minutes. The elaborate risk machinery recovered exactly what one categorical variable already gave. That is a negative result and it is reported as one.

12.7 PLS-gap gates

Variant	Held-out MAE	Gain
pls_gap_bin_gamma_positive	31.756141	+0.0141
risk_surface_pls_gamma_positive	31.681709	+0.0886
sched_x_pls_gap_gamma_positive	31.695244	+0.0750
family_x_pls_gap_gamma_positive	31.657513	+0.1128

The PLS disagreement geometry carries signal, but it destabilises under fine segmentation. The reliable question is not “*how large is the PLS gap?*” but “*how much should the PLS correction be trusted for this clinical family?*”

12.8 Where the risk gate helps and where it pays

Stratified by strict-final absolute error:

Error stratum	Gain
0–15 min	−0.4138
30–60 min	+0.6649
60–120 min	+0.3875
120+ min	+0.6924

Stratified by correction magnitude:

Absolute PLS correction	Gain	<i>n</i>
0–5 min	+0.0359	2,176
5–10 min	+0.4081	618
10–20 min	+2.9319	45

Two clean readings. The gate does what it was built to do on badly-predicted cases, and **pays for it by perturbing already-good ones** — a genuine trade, not a free lunch. And the upside concentrates almost entirely in the small subset where the note correction is non-trivial: 45 cases carry a +2.93 minute gain.

13. Stage 5 — the top-refine meta-gate (selected model)

13.1 Construction

The final model blends two already cross-fitted candidates.

	Candidate
A	pairblend_proc_risk_abs_et3_x_family_cap_gamma__risk
B	signed_resid_hgb8_x_family_cap_gamma

The **gate score** is the disagreement between two calibrated predictors:

$$d_i = \hat{y}_i^{\text{risk_hgb4}} - \hat{y}_i^{\text{procedure_family}}.$$

Using *disagreement between calibrated candidates* as the gate — rather than a raw risk score — is the key structural choice. Disagreement is a model-internal signal that requires no target, so it is clean by construction, and it localises exactly the cases where the trust question is live.

For candidate predictions a_i, b_i the pair blend is

$$\hat{y}_i(w) = (1 - w) a_i + w b_i,$$

with w searched over $\{0, 0.01, \dots, 1\}$ minimising fold-training MAE. Score-bin edges are quantiles of the fold-training scores only, with 6 bins, a minimum local sample size of 80, and a required local MAE improvement margin of 0.02 minutes before a local override is accepted.

13.2 Fitted policy

Learned score edges:

-16.7446, -1.2492, -0.3816, -0.0305, 0.0751, 0.6576, 16.2156

Score region	Weight on candidate B
Bin 0 — lowest disagreement	1.00
Bin 1	0.00
Bins 2, 3, 4	0.37
Bin 5 — highest disagreement	0.00
Global default	0.37

For OOF rows, edges and weights are always fitted without the validation fold. For held-out rows, the policy is refit on all training OOF evidence and applied unchanged.

Read the policy. It is not a monotone trust curve. At *extreme* disagreement in either direction (bins 1 and 5) the gate falls entirely back to candidate A — when the two calibrated models disagree violently, the more conservative one wins. At the *lowest* disagreement bin it goes entirely to candidate B. In the middle it blends. That shape is defensible, but with only four distinct fitted weights and a 0.02-minute acceptance margin it is also the most over-fittable component in the system, which is why its incremental gain has a confidence interval that crosses zero (Section 13.6).

13.3 Held-out performance by procedure family

From the stage-2 interpretability pass (base → strict PLS final, $n = 2,839$):

Family	n	Base MAE	Final MAE	Gain	Improved	Mean actual	Mean correction	Alignment
AF/PVI ablation	677	34.409	33.102	+1.307	61.30%	217.1	4.38	64.25%
Lead/device extraction-removal	76	49.141	48.415	+0.727	46.05%	216.5	2.44	48.68%
VT/PVC ablation	295	54.196	53.859	+0.337	53.22%	294.9	3.47	55.59%
Device implant	712	25.520	25.188	+0.332	52.67%	145.4	2.70	55.62%
Other ablation	365	36.598	36.295	+0.304	52.05%	200.1	3.63	54.25%
Other	115	19.070	18.907	+0.162	52.17%	64.8	2.40	53.91%

Family	n	Base MAE	Final MAE	Gain	Improved	Mean actual	Mean correction	Alignment
LAA closure	47	23.793	23.699	+0.094	55.32%	160.9	3.62	59.57%
Device revision-generator-upgrade	306	23.197	23.297	-0.099	47.06%	130.3	2.82	50.98%
Diagnostic / testing	233	23.668	23.907	-0.239	45.92%	107.1	3.23	48.50%

AF/PVI ablation carries the model. 677 cases, +1.307 minutes of gain, 61.30% improved, and an alignment rate of 64.25% — the only family where the correction points the right way appreciably more often than chance. Note also its mean base residual of -10.64 minutes: the tabular model systematically *over*-predicts AF/PVI, and the notes know it.

The two families with negative gain — device revision and diagnostic/testing — are exactly the ones Stage 3 subsequently assigned trust scalars of 0.775 and 0.300. The family calibration is repairing a failure the interpretability pass had already identified.

13.4 Latent block attribution

Manifold block	Mean contribution (min)	Mean signed contribution (min)
$u - t$ (directional)	8.302	-0.527
$ u - t $ (magnitude)	5.046	-0.022
$t \odot u$ (product)	2.768	-0.017

The directional block dominates, at roughly 1.6× the magnitude block and 3× the product block. The signed contributions are all slightly negative, consistent with the global correction mean of -2.17 minutes: **the note channel’s net effect is to shorten predictions.**

13.5 Concept-flag diagnostics

Concept flag	n	Prevalence	Gain when present	Gain when absent	Lift
congenital_complex_anatomy	241	8.49%	+0.4399	+0.4721	-0.0322
access_support	377	13.28%	+0.2670	+0.5004	-0.2333
vt_pvc_instability	1,157	40.75%	+0.2210	+0.6402	-0.4192
redo_prior_ablation	442	15.57%	+0.2180	+0.5157	-0.2977
device_lead_complexity	345	12.15%	+0.1435	+0.5144	-0.3709

Every lift is negative. Cases where a concept flag fires improve *less* than cases where it does not. vt_pvc_instability fires on 40.75% of cases and has the worst lift of all.

This is the empirical case for Section 8 stated in a table. The flags are not merely weak — they mark the cases the model handles *worse*. Whether that is because the flags are miscalibrated or because flagged cases are intrinsically harder cannot be separated with a binary indicator, and that ambiguity is itself the argument for graded, assertion-aware extraction.

13.6 Selected model metrics

Metric	Value
Train OOF MAE	32.7156839203
Held-out MAE	31.5564180731
Gain vs strict PLS	+0.2138455833
95% paired bootstrap CI	[+0.1001, +0.3259]
Gain vs procedure-family gate	+0.0511042965
95% paired bootstrap CI	[-0.0108, +0.1123]
Improved-case rate vs strict PLS	48.89%
Large wins (> 10 min)	23
Large losses (< -10 min)	16

The honest reading. The gain over the strict PLS manifold is solid: the interval excludes zero. The gain of the meta-gate *over the simple procedure-family gate* is 0.051 minutes with an interval that **crosses zero**. The final two stages of the ladder are not individually established. Note also that the improved-case rate is 48.89% — below half — meaning the gate wins by making its wins larger, not more frequent.

13.7 Model-selection rule and the lockbox candidate

The reported model is the candidate with the best **training OOF** MAE in the targeted top-refine search. The search also recorded a better held-out candidate:

	Selected	Exploratory
Variant	topref_pair__pairblend_proc_...topref_pair__pairblend_proc_...	topref_pair__pairblend_proc_...topref_pair__pairblend_proc_...
Train OOF MAE	32.7156839203	32.7806355030
Held-out MAE	31.5564180731	31.5234029457
Selected?	yes	no — worse train OOF

The exploratory candidate is **not** the claim. It is held as a lockbox candidate only. Choosing it because it scored better on held-out would convert the held-out set into a training set.

14. The complete algorithm

INPUT

```
tabular case table
timestamped clinical-note table
prediction-timestamp definition
```

— BOUNDARY —

- 1 drop rows with invalid schedule time or non-positive target
- 2 sort cases by scheduled_in_room
- 3 cut at the 80th percentile -> candidate held-out cohort
- 4 remove every held-out patient from the earlier training rows
- 5 attach to each case only notes with doc_date < scheduled_in_room - 2 days
- 6 reject post-operative doc types; cap note count / length / lookback

— TABULAR FOUNDATION —

- 7 build planned-schedule, case, patient and comorbidity features
- 8 in patient-grouped folds:
 - fit preprocessing + procedure/surgeon target encodings on fold-train

```

    fit xgb_tuned_v2 and xgb_absoluteerror_same_shape
    predict fold-validation -> base_00F
9  refit both on all training rows, predict held-out -> base test
10 base = 0.60 * squared-error member + 0.40 * absolute-error member

```

— NOTE / TABULAR MANIFOLD —

```

11 residual target r = y - base_00F
12 fit note TF-IDF -> SVD(160) -> scale (train only)
    fit tabular SVD(96) -> scale, keep first 64 (train only)
13 fit PLS(40) between the two views; slice to 26 components
14 build delta geometry phi = [u-t, |u-t|, t*u] -> 78 features
15 fit Huber residual head in patient-grouped folds -> r_hat
16 select gamma on train 00F MAE over {0.000 ... 1.600} -> 1.550
    y_PLS = base + gamma * r_hat

```

— CONDITIONAL TRUST —

```

17 learn procedure-family trust gamma_f inside outer patient folds
    y_family = base + gamma_f * (y_PLS - base)
18 build nested risk / signed-residual / gain candidates
    every score model trained without its own validation fold
    every bin edge and cap from outer-training rows only
19 cross-fit the six-bin pair-blend policy on candidate disagreement

```

— SELECTION AND SCORING —

```

20 select the top-refine variant by TRAIN 00F MAE
21 refit all training-derived transforms and policies on all training rows
22 apply once to held-out; compute held-out MAE

```

Every learned object in this list — imputation medians, category frequencies, target encodings, TF-IDF vocabulary, SVD bases, scalars, PLS directions, residual head, γ , family trust scalars, risk models, bin edges, caps, blend weights — is fit on a fitting partition and applied to an evaluation partition. There are no exceptions.

15. Iteration: one pass completed, convergence not reached

This is the most important open section of the document. Everything above describes a *single* residual iteration. The architecture is a truncated instance of a well-defined iterative scheme, and the truncation was a decision about effort, not a statement about convergence.

15.1 The iteration operator

Let $p_i^{(0)}$ be the current out-of-fold prediction for case i . At stage k define the residual

$$r_i^{(k)} = y_i - p_i^{(k)},$$

train a new head h_k on $(x_i, z_i, p_i^{(k)}) \mapsto r_i^{(k)}$ under nested cross-fitting, and update

$$p_i^{(k+1)} = p_i^{(k)} + \eta_k h_k(x_i, z_i, p_i^{(k)})$$

with shrinkage $\eta_k \in (0, \eta_{\max}]$ selected **inside training folds**.

What we have executed is exactly two applications of this operator with a conditional trust gate attached:

k	$p^{(k)}$	h_k	η_k
0	tabular base	PLS manifold Huber head	$\gamma = 1.550$
1	strict PLS model	family / risk / meta-gate re-weighting	γ_f , blend w
2	not run	—	—

Stage $k = 1$ is not a full residual head — it is a *re-weighting* of the stage-0 correction rather than a new learner on the stage-1 residual. **A genuine second residual iteration has never been attempted.**

15.2 The functional-gradient view

The scheme is gradient boosting in function space under absolute loss. For $\ell(y, f) = |y - f|$,

$$-\frac{\partial \ell}{\partial f} = \text{sign}(y - f),$$

so the exact functional gradient step fits the **sign** of the residual, not its magnitude. Our heads fit the residual itself under a Huber loss, which interpolates: quadratic near zero (magnitude-sensitive, efficient) and linear in the tail (sign-like, robust). The learned scalar γ then plays the role of the line-search step length that a pure gradient step would require.

This framing has one immediate and uncomfortable consequence. **Section 12.4 showed that residual direction is essentially unlearnable from our features** — OOF correlation with signed residual is 0.079. The exact L_1 functional gradient is a pure direction object. So the honest statement is:

The reason further iterations have not obviously succeeded is that the natural gradient direction for absolute loss is the one signal our feature set cannot predict.

This is not a reason to stop. It is a precise specification of what iteration 2 must supply: **a direction signal**, not another magnitude model.

15.3 Why naive iteration fails, and the nesting requirement

If h_k is trained on residuals from a prediction that saw the row’s own label, then h_k learns the *memorisation* of $p^{(k)}$, not its *error*. Applied out of sample, that correction is noise with the sign of overfit — actively harmful.

The requirement is therefore absolute:

Every input to stage $k + 1$ must be produced without the row’s own target.

Concretely, at each iteration:

1. All upstream candidate predictions are generated **inside** the outer loop.
2. Risk labels, score bins, blend weights, caps and residual scalars are fit on outer-fold training rows only.
3. Minimum region sizes and global fallbacks are enforced.
4. The full candidate set is frozen before any new lockbox is opened.

The cost is multiplicative: iteration k requires all of iterations $0..k - 1$ regenerated within the outer loop. This is the real reason only one iteration has been completed.

15.4 Does the iteration converge?

Write $\mathcal{R}(p) = \mathbb{E}|Y - p|$. Under exact line search and a non-degenerate head, each step is non-increasing:

$$\mathcal{R}(p^{(k+1)}) \leq \mathcal{R}(p^{(k)}),$$

so the population risk sequence is monotone and bounded below by R^* , hence convergent. Three caveats stand between that statement and practice.

Convergence is to a restricted optimum. The limit is the best predictor in the closure of the span of the head class, not R^* . If the head class cannot represent the remaining structure, the sequence converges above the floor and the gap is approximation error, not noise.

We observe $\widehat{\mathcal{R}}$, not $\mathcal{R}$. The empirical OOF risk has standard error of order $\text{sd}(|e|)/\sqrt{n}$; with $n \approx 10,900$ and heavy-tailed errors this is roughly 0.3–0.4 minutes. **Stage gains of 0.05 minutes are far inside that noise.** The monotonicity that holds in population does not hold in the sample, and our own trace shows it: `crossfit_residual_hgb` won on train OOF and lost on held-out.

Selection consumes information. Each iteration chooses among candidates using the same training OOF estimate, so the estimate degrades as a selector with every pass. This is the winner’s-curse channel, and it compounds.

15.5 Stopping rules

Four rules, to be applied jointly, not alternatively. Iteration stops when **any** fires.

S1 — Floor gap. Stop when the gap between the achieved risk and the estimated attainable floor is smaller than the resolution of the evaluation:

$$\widehat{\mathcal{R}}(p^{(k)}) - \widehat{E} < \epsilon,$$

where $\widehat{E}$ estimates the attainable floor. This is the only rule that references the problem rather than the procedure, and it is the one worth investing in.

S2 — Confidence. Stop when the lower bound of a patient-clustered bootstrap interval on the OOF improvement is not positive:

$$\text{LB}_{95\%} \left[\widehat{\mathcal{R}}(p^{(k)}) - \widehat{\mathcal{R}}(p^{(k+1)}) \right] \leq 0.$$

By this rule the current iteration has already stopped at stage 5. The gain of the meta-gate over the procedure-family gate is +0.0511 with interval $[-0.0108, +0.1123]$. S2 says: do not add stage 5 without new evidence.

S3 — Operational negligibility. Stop when the gain is real but too small to matter. A gain below roughly 0.1 minutes per case is invisible against OR scheduling granularity. The right threshold is a business decision, stated in minutes of OR time per year, not in MAE.

S4 — No progress / budget. Stop after k consecutive iterations in which no candidate passes S2, or when the compute budget for nested regeneration is exhausted.

15.6 Connecting the stop rule to the irreducible floor

S1 is only meaningful with an estimate of the floor. Parallel work on this same cohort — the Peras irreducible-risk framework, on a closely related but **not identical** pipeline ($n = 12,993$, different exclusions and feature arms) — produces:

Quantity	Value	Interval
Certified lower bound B^*	28.85 min	≥ 28.27 at the 5% quantile
Conservative proved rung	24.18 min	≥ 23.60
Estimated attainable floor $\hat{E}$	31.12 min	[30.40, 31.78]

Read against the present model at 31.5564:

- Distance to the **certified** floor: $31.5564 - 28.85 = 2.71$ minutes. This is the absolute ceiling on what any further modelling could win on this information set.
- Distance to the **estimated attainable** floor: $31.5564 - 31.12 \approx 0.44$ minutes, and 31.5564 sits inside $\hat{E}$'s interval [30.40, 31.78].

The operational implication is blunt. Under S1 with ϵ set anywhere near the evaluation's own resolution, **this model is already at the estimated attainable floor for its information set.** Further iteration on the same features is not expected to recover more than a few tenths of a minute.

Two caveats, both important. The floor estimates come from a different pipeline and are not directly transferable without recomputation on this exact feature matrix. And the floor is a property of **the information set**, not of the method — which points at the correct move.

15.7 What iteration 2 should actually be

Given S1 and Section 15.2, the conclusion is not *iterate the same scheme harder*. It is:

Change the information set, not the estimator.

R^* moves only when the representation changes. Everything else — better heads, more bins, finer gates — moves how tightly we can approach a fixed floor, and we are already close to it.

Ranked by expected value:

1. **Replace the narrative extraction layer (Section 8).** This is the only proposed change that alters R^* itself. Graded, negation-aware, assertion-level concept evidence is a genuinely different information set from five saturated regex flags.
2. **Recompute the floor on this exact pipeline.** Without it, S1 cannot be applied honestly. This is cheap and should come first in wall-clock order.
3. **Attack the direction problem.** Iteration 2 needs a signed-residual signal, not another magnitude model. Candidate sources: note recency distribution, contradiction between planned procedure and narrative context, candidate spread, calibrated uncertainty, pre-cutoff patient-history summaries.
4. **Nested forward-time selection.** Replace patient-only GroupKFold with nested forward-time folds with patient purging, so the selection estimate matches the deployment question.
5. **Open a fresh lockbox.** Freeze everything, and evaluate once on a later untouched cohort.

15.8 Per-iteration diagnostics to track

Every future iteration should report, before any accept/reject decision:

Diagnostic	Why
Train OOF MAE and its clustered bootstrap interval	S2
Correction magnitude: mean, median, mean absolute	drift detection
Alignment rate $\Pr[\text{sign}(\hat{r}) = \text{sign}(r)]$	direction quality (currently 56.3%)
Correlation of correction with true residual	currently 0.158
Per-family gain, including negative families	harm detection

Diagnostic	Why
Per-coverage-stratum gain	note availability differs across splits
Large-win and large-loss counts	tail behaviour
Improved-case rate	currently 48.89% — below half
Distance to $\hat{E}$ and to B^*	S1

The alignment rate deserves emphasis: at 56.3% the correction points the right way barely more often than a coin. That number, more than any MAE figure, says how much room the direction channel still has.

16. Validation and leakage audit

16.1 Controls in force

Control	Status	Evidence
Later temporal held-out cohort	yes	cut at the 80th percentile of <code>scheduled_in_room</code>
Zero patient overlap	yes	$\mathcal{P}_{\text{train}} \cap \mathcal{P}_{\text{held-out}} = \emptyset$, audited
Outcome-time columns excluded	yes	all <code>case_actual_*</code> removed from the feature frame
Exact note timestamps with a 2-day buffer	yes	295,463 dates checked, 0 violations, min margin 2.5 h
Post-operative doc types rejected	yes	145,426 rows rejected
Same-day pre-operative notes disabled	yes	cache setting
Fold-local target encoding	yes	procedure and surgeon, fit on fold-train only
OOF base predictions used for residual targets	yes	never in-sample
Patient-grouped residual and gate evaluation	yes	GroupKFold(5) on patient
Nested construction of target-derived risk scores	yes	score model trained without its own validation fold
Bin edges and blend weights from outer-training only	yes	meta-gate policy
Selection by train OOF, not by held-out	yes	31.5234 candidate rejected
Forbidden 11-feature LLM file not read	yes	<code>forbidden_llm_feature_file_read = false</code> , 0 access-like hits
Held-out target used only for final scoring	yes	
Saved prediction artefact reproduces the metric	yes	31.556418073067334

16.2 Audit values

```
train rows:                10,882
held-out rows:             2,839
train patients:            8,948
held-out patients:         2,499
patient overlap:           0
missing patient ids:       0 / 0
note dates checked:        295,463
cutoff violations:         0
smallest margin before cutoff: 2.5 hours
forbidden LLM feature file read: false
access-like hits to that file: 0
```

The forbidden-file scan found the filename in exactly two places, both defensive:

```
strict_manifold_text_sweep.py:52  FORBIDDEN = PROJECT/"data"/"mayo"/"mayo_hrs_llm_features_top11.
strict_manifold_text_sweep.py:72  assert FORBIDDEN.name == "mayo_hrs_llm_features_top11.parquet"
```

17. Limitations

Stated plainly, in descending order of how much they should worry a reader.

The held-out cohort was viewed repeatedly. The strict PLS run alone evaluated 127 candidate variants, and several subsequent passes examined the same held-out set. Selection was by train OOF, which is materially better than picking the best held-out row, but repeated examination still influences research decisions. **31.5564 is a strong retrospective held-out estimate, not publication-grade independent confirmation.** The only fix is a fresh lockbox.

The last two rungs of the ladder are not individually established. The meta-gate's gain over the simple procedure-family gate is +0.0511 with a bootstrap interval that crosses zero. The defensible claim stops at the procedure-family gate (31.6075, CI [+0.0565, +0.2664] over strict PLS). Everything after that is promising, not proven.

Internal folds are patient-disjoint but not chronological. They control the dominant contamination risk and do not simulate forward-time deployment. Nested forward-time folds with patient purging would give a better estimate of future performance.

The reported interval is a paired case-level bootstrap. It does not fully account for repeated-patient correlation. A patient-clustered bootstrap is preferred and was used for the stage-2 figures; the later-stage intervals should be recomputed on that basis.

Scheduling-derived features change the task. They encode operational planning expertise and improve accuracy, but their availability and stability at inference must be guaranteed, and the revision/feedback risks of Section 5.4 must be audited.

Note coverage differs between splits (92.05% train vs 99.79% held-out). Monitoring must stratify by note availability and volume; an aggregate gain can hide a coverage-driven artefact.

The concept-flag block has negative gain lift on every flag (Section 13.5). The narrative extraction layer is the least defensible component in the system.

Alignment rate is 56.3%. The correction's direction is barely better than chance. The model wins on magnitude allocation, not on knowing which way to move.

The package contains source and predictions, not fitted estimators. Reproducing training requires the authorised raw data and the pinned environment.

18. Deployment contract

At inference. Provide the same tabular schema and planned schedule fields, plus only notes finalised before the two-day cutoff. Preserve training-time category handling and transformations. **Do not** recompute target encodings, TF-IDF vocabulary, SVD bases, PLS directions, score-bin edges or blend weights using production outcomes.

Fallback behaviour. If the note bundle is empty or the extraction service is unavailable, fall back to the tabular base and record a degradation flag. Never silently change the input distribution.

For retraining. Create a new chronological cutoff, rebuild note bundles, regenerate all OOF candidate predictions inside the outer loop, reselect using training-side validation only, freeze the model, and then evaluate once on a later untouched cohort.

Monitoring. Stratify by procedure family, note coverage, note volume, and scheduled-duration bin. Track correction magnitude and alignment rate as drift indicators — they move before MAE does.

19. Reproduction

Install from requirements.txt; configure the authorised clinical-data repository via CLINICAL_NOTES_PROJECT. The original run order is in scripts/run_best_31_5564_pipeline.sh.

Step	Script
1	src/build_raw_strict_high_signal_split.py
2	src/validate_strict_raw_run.py
3	src/tabular_model_sweep.py
4	src/strict_manifold_push.py
5	src/risk_category_modulation.py
6	src/variance_aware_pls_modulation.py
7	src/strict_risk_family_k_sweep.py
8	src/strict_meta_gate_push.py
9	src/strict_meta_gate_top_refine.py

To validate the distributed result **without** raw data:

```
python scripts/verify_saved_result.py
```

Expected:

```
heldout_rows=2839
heldout_mae=31.556418073067334
status=PASS
```

The archive excludes raw clinical data, note text, fitted binaries, and credentials.

20. Notation

Symbol	Meaning
y_i	realised procedure duration, minutes
x_i	structured (tabular) covariates
z_i	eligible pre-operative note bundle
τ_i	prediction timestamp (case scheduling moment)
$\mathcal{F}_t$	σ -field of everything knowable at time t
$\hat{y}^{\text{base}}$	60/40 tabular blend

Symbol	Meaning
r_i	residual target, $y_i - \hat{y}_i^{\text{base, OOF}}$
X, Z	tabular and note latent views entering PLS
t_i, u_i	PLS scores, tabular and note, $\in \mathbb{R}^{26}$
ϕ_i	delta-geometry feature vector, $\in \mathbb{R}^{78}$
$\hat{r}_i$	residual head output
γ	global trust scalar, 1.550
γ_f	procedure-family trust scalar
C_c	correction cap for category c
d_i	meta-gate score (candidate disagreement)
w	pair-blend weight on candidate B, 0.37 global
$p^{(k)}$	prediction after k residual iterations
η_k	iteration shrinkage
R^*	irreducible risk on the information set
$B^*, \hat{E}$	certified floor, estimated attainable floor
$\odot$	elementwise (Hadamard) product

21. Roadmap

#	Action	Changes R^* ?	Effort	Priority
1	Recompute the irreducible floor on this exact pipeline	no	low	first
2	Replace regex concepts with retrieval + rerank + service extraction (§8)	yes	medium	highest value
3	Build a direction signal for iteration 2 (§15.7)	partially	medium	high
4	Nested forward-time selection with patient purging	no	medium	high
5	Re-run later-stage intervals as patient-clustered	no	low	medium
6	Audit scheduling-field revision risk	no	low	medium

#	Action	Changes R^* ?	Effort	Priority
7	Freeze and open a fresh lockbox	no	low	required before any external claim
8	Per-case uncertainty via multi-quantile head + conformal calibration	no	medium	medium

22. Final statement

The Conditional Residual Clinical Notes Learner reaches a held-out MAE of **31.5564180731 minutes** on 2,839 later cases from patients absent from training, using structured clinical and planned scheduling data together with clinical notes restricted to text finalised more than two days before each case. It does not use the former 11-variable LLM feature table.

Of the 0.683 minutes gained over the tabular base, 0.469 minutes — the majority, and the only component whose interval comfortably excludes zero — comes from the clinical narrative entering as a residual correction through a PLS shared manifold. The conditional trust machinery above it is coherent, clinically interpretable, and individually unproven at its last two rungs.

One residual iteration has been completed. The iteration operator, the shrinkage schedule, the nesting requirement and four stopping rules are specified in Section 15. Under the confidence rule S2 the current ladder has already stopped; under the floor-gap rule S1, and using a parallel estimate of the attainable floor, this model sits roughly 0.44 minutes above the estimated floor for its information set and about 2.71 minutes above the certified one.

That arithmetic is the whole argument for what comes next. Further iteration on the same features is close to exhausted. **The remaining headroom is in the information set, not in the estimator** — which is why the highest-value next step is not another gate, but the replacement of five regular expressions with a retrieval-ranked, reranked, assertion-aware extraction layer, followed by a single frozen evaluation on a cohort no one has seen.

Conditional Residual Clinical Notes Learner — Itamar Zernitsky, Department of Mathematics, Bar-Ilan University; Opmed.ai informal “Shadows” research team. Prepared for Opmed.ai.